

A Hybrid End-to-End Neural Network for Accurate Global Natural Gas Trade Forecast

Yu Yan, Gang Lu, Xiaoqing Yan, Peng Xia, Jia Chen, *Member, IEEE*, Jie Yang, and Di Wu , *Member, IEEE*

Abstract—Accurate forecast of global natural gas trade is crucial for a country to promptly optimize and adjust its energy strategies for energy security. Recently, a variety of machine learning-based approaches have been adopted to address this issue. However, global natural gas trade is influenced by various complex (e.g., reserves of natural gas, energy imports and exports, economy, population, etc.) and uncertain factors (e.g., climate change, international turmoil, and global pandemics). Existing approaches either partially or loosely explore these factors by nonend-to-end learning, resulting in suboptimal feature extraction and temporal integration for forecasting. To fill this gap, this article proposes a novel hybrid end-to-end neural network (HENN) with three components: graph attentional network (GAT), temporal transformer (TT), and multilayer perceptron (MLP). First, GAT learns the latent features of annual trade network among various countries year by year. Second, TT integrates these annual latent features with temporal information to forecast the trade network's representations for the upcoming year. Third, MLP employs the forecasted representations to forecast the trade network of next year. The three components are trained by end-to-end learning, ensuring the feature extraction and the integration of temporal information are both comprehensive and optimal for forecasting. Extensive experiments with global natural gas trade data from 2000 to 2023 show that the proposed HENN significantly outperforms advanced models in forecasting the global natural gas trade.

Index Terms—Global natural gas trade forecast, graph attentional network (GAT), multilayer perceptron (MLP), transformer.

NOMENCLATURE

Notation	Description
G^t	Static graph at t moment.
V	Node set.
E^t	Weight matrix at t moment.
F	Number of features of each node.
H^t	Feature matrix at t moment.
A^t	Adjacency matrix at t moment.
hidden_dim	Length of the vector representation of each node.
threshold	Criteria for selecting features in PCC.
layers	Layer numbers of MLP.
heads	Attention head numbers of TT.
X^t	Vector representation of graph at t moment.
X_u^t	Vector representation of node u at t moment.
e_{uv}^t	Vector representation of edge between node u and node v at t moment.
S^t	Vector representation of static graph at t moment.

I. INTRODUCTION

ENERGY is vital for sustainable development, influencing a nation's social stability, economic growth, and environmental health. Over the past two decades, the global demand for energy has witnessed a dramatic upsurge [1]. However, the current energy landscape is largely dominated by traditional fossil fuels (e.g., coal and oil), which have triggered pressing environmental challenges, including escalating greenhouse gas emissions and accelerated global warming [2]. Natural gas emerges as a viable and rational alternative to these fossil fuels, primarily due to two reasons: 1) it is cleaner than oil and coal and 2) it is more cost-effective than many renewable energy sources [3]. As a result, natural gas has become a cornerstone of national energy security. Accurate global natural gas trade forecast is also indispensable for the reasons given below: First, it supports businesses and government agencies in crafting long-term energy strategies and policies. Second, it helps nations prevent energy shortages or surpluses, reducing societal strain. Third, by forecasting natural gas trade, it enables effective planning and management of energy supplies, ensuring stability and security [4].

The rapid advancement of machine learning (ML) has driven the development of diverse ML-based approaches for forecasting global natural gas trade [5]. Poczeta and Papageorgiou et al. [6] introduced a novel hybrid computational intelligence model that seamlessly integrates an evolutionary learning-driven fuzzy

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cognitive map with a conventional artificial neural network. Sankar et al. [7] proposed the dynamic self-attention network (DySAT), a novel neural architecture crafted to capture node representations reflecting the dynamics of evolving graph structures. Pareja et al. [8] proposed EvolveGCN, which adopts the graph convolutional network (GCN) model along the temporal dimension without resorting to node embeddings. Rossi et al. [9] presented temporal graph networks (TGN), a generic, effective framework for deep learning on dynamic graphs represented as sequences of timed events. Xu et al. [10] proposed the temporal graph attention (TGAT) layer to efficiently aggregate temporal-topological neighborhood features as well as learn the time-feature interactions. Further advancing the field, Chen et al. presented the GC-LSTM model, and Ma et al. proposed a novel algorithm by joint multilabel learning and feature extraction (MLjFE), capable of extracting both local and global insights concurrently [11], [12]. Most recently, Tian et al. [13] unveiled the frequency enhanced continuous-time dynamic graph (FreeDyG) model, which facilitates a deeper understanding of interaction patterns by capturing both periodic and shifting behaviors. However, these methods adopt nonend-to-end learning frameworks, either partially or loosely exploring complex and uncertain factors in global natural gas trade, which results in suboptimal feature extraction and temporal integration for forecasting.

To address this challenge, this article proposes a novel hybrid end-to-end neural network (HENN) with three components: graph attentional network (GAT), temporal transformer (TT), and multilayer perceptron (MLP). First, GAT learns the latent features of annual trade network among various countries year by year [14]. Second, TT integrates these annual latent features with temporal information to forecast the trade network's representations for the upcoming year [15]. Third, MLP employs the forecasted representations to forecast the trade network of next year [16]. The three components are trained by end-to-end learning, ensuring the feature extraction and the integration of temporal information are both comprehensive and optimal for forecasting. HENN significantly differs from existing approaches as follows.

- 1) HENN adopts an edge-centric temporal forecasting formulation, directly modeling the evolution of edge representations and predicting the entire next-year adjacency matrix. In contrast, models such as DySAT and GCN_MA primarily follow a node-centric paradigm, where temporally-aware node embeddings are first learned and then reused for downstream link prediction.
- 2) TT is applied directly to sequences of edge representations to predict future network states, rather than merely encoding historical dynamics. This distinguishes HENN from DySAT-like approaches, where temporal attention is mainly used for representation learning rather than future network inference.
- 3) HENN is trained in a fully end-to-end manner, jointly optimizing spatial aggregation, temporal reasoning, and network-level prediction toward the next-year adjacency matrix.

The essential contributions of this article are delineated in the following.

- 1) Next-year global natural gas trade network forecasting is formulated under yearly snapshot dynamics. Besides,

an edge-centric temporal modeling route is constructed to directly predict the next-year adjacency matrix of the global natural gas trade network in an end-to-end manner.

- 2) An advanced and reliable HENN model is proposed for forecasting global natural gas trade networks. Its principles, technical details, and algorithm designs are also provided.
- 3) Extensive empirical studies are conducted on the collected dataset for evaluating the proposed HENN model, including comparisons with advanced models, ablation experiments, and parameter sensitivity analyses.

The experimental results demonstrate that the proposed HENN model significantly outperforms current advanced models in forecasting the global natural gas trade. The source code of the proposed HENN model and the collected datasets are, respectively, shared at the anonymous links.¹ The rest of this article is organized as follows. Section II discusses the related work. Section III introduce the preliminaries. Section IV presents the proposed method. Section V provides and discusses the experimental results. Finally, Section VI concludes this article.

II. RELATED WORK

A. Traditional Forecast Methods

Traditional forecast methods encompass a variety of techniques, including time series analysis [17], regression analysis [18], Monte Carlo simulation [19], system dynamics model [20], and methods based on expert judgment [21]. Time series analysis, one of the earliest and most foundational methodologies, leverages historical data to identify patterns, trends, and seasonal variations. By applying statistical modeling and curve-fitting algorithms, it generates reliable predictions for future energy demand or production [17]. Regression analysis offers a powerful statistical framework to examine relationships between a dependent variable, such as energy consumption, and independent variables like economic growth, weather patterns, and population changes [18]. Monte Carlo simulation, a probabilistic approach, uses random sampling to model a wide range of potential outcomes, making it particularly effective in addressing uncertainties in variables, such as fuel prices, demand variability, and production capacity [19]. System dynamics model takes a comprehensive approach, simulating the intricate interactions of internal and external factors within the energy ecosystem, including policy shifts and technological advancements [20]. Methods based on expert judgment remain a valuable tool in energy forecasting. Subjective expert assessments are collected through interviews, surveys, and expert group meetings and then integrated to form forecasts that take advantage of the collective wisdom and experience of these experts [21]. Despite their strengths, these traditional methods face significant challenges in capturing the complexities of global natural gas trade. Complex factors (e.g., reserves of natural gas, energy imports and exports, economy, population, etc.) and various uncertain factors (e.g., climate change, international turmoil, and global pandemics) introduce levels of uncertainty that these methods struggle to capture.

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B. ML-Based Forecast Methods

The rapid evolution of ML has spurred a growing number of studies leveraging ML techniques to tackle the complex challenge of forecasting global natural gas trade. Zeng et al. [22] introduced the Verhulst model, a significant advancement in time series analysis, which introduced a novel time response function alongside an optimized initial value method, addressing limitations in traditional time series models by better capturing nonlinear growth patterns. Du et al. [23] proposed a novel deep learning prediction method, integrating knowledge-enhanced graph-based temporal-spatial networks, enabling the model to effectively capture temporal and spatial relationships in natural gas consumption data. Qiao et al. [24] introduced an improved model combining improved sparrow search algorithm (ISSA), long short-term memory (LSTM), and wavelet transform (WT), in which WT is used to decompose the natural gas consumption series, LSTM performs component-wise forecasting, and an ISSA optimizes LSTM hyperparameters, with final predictions obtained by reconstruction. Fragalla et al. [25] proposed a hybrid multilevel discrete wavelet decomposition–transformer framework, where multilevel discrete wavelet decomposition stabilizes the gas production series, and an encoder–decoder transformer with multihead attention performs sequence-to-sequence forecasting. Despite these advancements, a critical gap persists in the application of ML-based models to global natural gas trade forecast. Existing models primarily focus on predicting natural gas demand, production, and consumption at regional or national scales [26]. They excel in analyzing localized patterns but struggle to capture the complex dynamics of international trade networks.

C. Methods for Temporal Link Prediction

A growing number of models, originally designed for traditional link prediction tasks, have been adapted to forecast global natural gas trade [27]. Lei et al. [28] introduced GCN-GAN model, which addresses the challenges of temporal link prediction in weighted dynamic networks by combining GCNs, LSTM units, and GANs [29], [30]. Sankar et al. [7] proposed the DySAT, a novel neural architecture employs self-attention mechanisms to model temporal changes in graph topology, enabling it to adaptively focus on relevant structural patterns over time. That same year, Xu et al. [10] developed TGAT layer, a robust tool that integrates temporal and topological neighborhood features while modeling complex time-feature interactions through attention mechanisms. Pareja et al. [8] proposed EvolveGCN, which captures the dynamism of the graphs sequence through using a recurrent neural networks (RNN) to evolve the GCN parameter. Two architectures are considered for the parameter evolution. Rossi et al. [9] presented TGN, which is a novel combination of memory modules and graph-based operators, a generic, effective framework for deep learning on dynamic graphs represented as sequences of timed events. Chen et al. presented the GC-LSTM model, a hybrid model that takes advantage of GCNs and LSTMs to simultaneously capture spatial relationships in graph structures and temporal patterns in time series data, offering a comprehensive approach to prediction of link dynamics [11]. Also Ma et al. [12] unveiled MLjFE model, an innovative framework for temporal link prediction that employs joint MLjFE to extract critical insights from the parameter matrix. Concurrently, Shao et al. [31] proposed

FastSTLSG model, a model that automatically analyzes topological features within a unified framework, effectively capturing spatio-temporal correlations in mobile ad hoc networks. Tian et al. [13] introduced the FreeDyG model, which provides deeper insights into interaction patterns by modeling both periodic and evolving behaviors. In the same year, Mei et al. [32] proposed the GCN_MA model, which combines GCNs, RNNs, and multihead attention to produce precise and detailed node embeddings. However, these models mentioned above adopt nonend-to-end learning frameworks, either partially addressing or loosely weighting these complex and uncertain factors, which results in suboptimal feature extraction and integration of temporal information, hindering their ability to fully capture the nuanced and interconnected dynamics of global natural gas trade.

III. PRELIMINARIES

A. Notations and Descriptions

The adopted notations and their descriptions in this article are summarized in the Nomenclature.

B. Dynamic Network Modeling

To use our model for precise forecast of global natural gas trade, a rigorous and systematic approach to modeling energy data is essential. The approach unfolds through three key stages: country identification, network construction, and feature characterization. Initially, we assign unique numerical identifiers to each country, for example, designating “China” as 1, “Japan” as 2, and so forth, to ensure consistency and facilitate analytical processes. Next, we construct annual trade networks where each static graph captures natural gas trade relationships between countries for a given year. For instance, in 2020, China and Japan are represented as nodes 1 and 2, respectively, with the edge weight between them reflecting the trade volumes conducted that year. In addition, each country is characterized by a range of features that significantly influence trade dynamics, such as natural gas reserves, energy imports and exports, economic indicators, and population metrics. We develop comprehensive node feature vectors that encapsulate these diverse attributes. Through this structured methodology, we model the complex dynamics of global natural gas trade with both precision and analytical depth.

C. Network Representation Learning (NRL)

NRL, also known as network embedding, encompasses a suite of techniques designed to transform nodes, edges, or subgraphs of a network into low-dimensional vector spaces while preserving the network’s structural, relational, and semantic properties [33]. By converting discrete, high-dimensional network data into continuous, compact vector representations, NRL enables efficient downstream ML tasks, such as node classification, link prediction, community detection, and network visualization. Key NRL methodologies include matrix factorization, which decomposes adjacency matrices to capture latent features; random walk-based approaches, which leverage node transition probabilities to encode proximity; and advanced deep learning frameworks, which model intricate network patterns. Among these, GNNs have emerged as a powerful paradigm for graph-structured data, distinguished by their capacity to

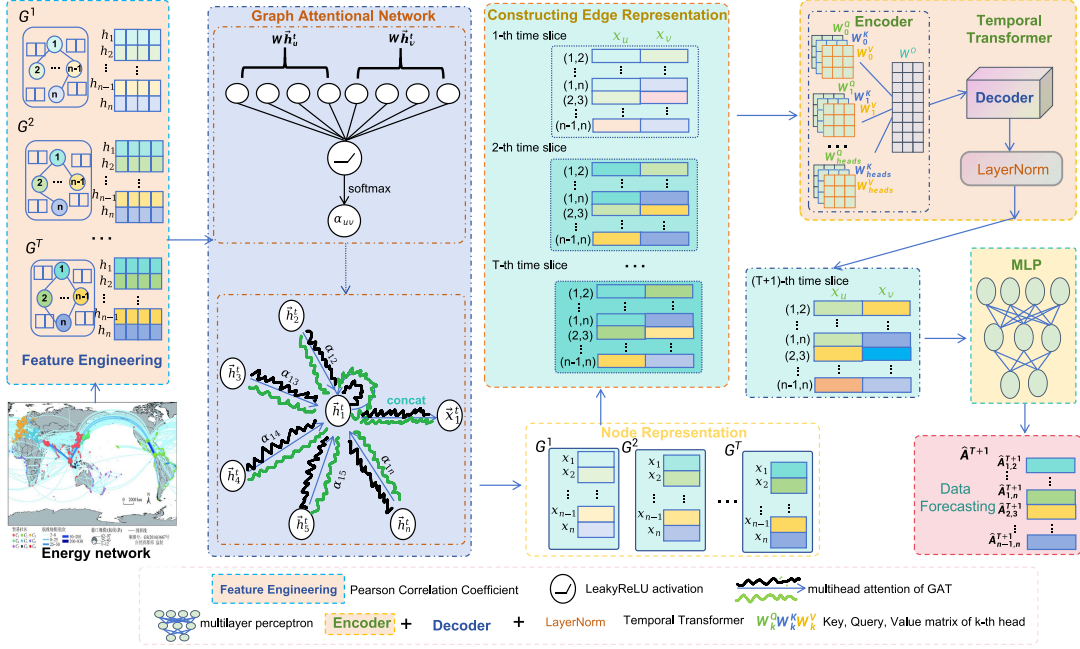


Fig. 1. Architecture of HENN.

capture complex topological patterns [34]. GNNs operate by iteratively refining node representations through the aggregation of information from neighboring nodes across multiple layers, seamlessly integrating local and global structural details with node and edge attributes. This iterative process enables GNNs to model both static and evolving network characteristics, making them particularly well-suited for applications in dynamic and heterogeneous networks, such as global trade systems, social networks, and biological interaction graphs. Recent advancements in NRL have introduced hybrid approaches that combine GNNs with temporal modeling techniques, such as LSTM units and attention mechanisms, to better capture the evolving nature of network structures over time. These temporal-aware NRL methods are particularly valuable in domains like global natural gas trade forecasting, where trade relationships exhibit both spatial dependencies and temporal dynamics.

IV. PROPOSED METHOD

A. Problem Formulation

A dynamic network can be characterized as a series of graph snapshots denoted by $G = \{G^1, G^2, \dots, G^T\}$, where each snapshot $G^t = (V, E^t, H^t)$ represents the network's state at a specific time point t ($t \in \{1, 2, \dots, T\}$).

For the graph snapshot at time t , we employ an adjacency matrix $A^t \in \mathbb{R}^{|V| \times |V|}$ to represent the static topological structure of the graph G^t . In our experiment, we define two threshold points: l and r , to categorize the edge weights. Specifically, for nodes i and j in the node set V , we assign $A_{ij}^t = 0$ if the edge weight $E_{ij}^t < l$, $A_{ij}^t = 1$ if $l \leq E_{ij}^t \leq r$, and $A_{ij}^t = 2$ if $E_{ij}^t > r$. In addition, the feature matrix $H^t \in \mathbb{R}^{|V| \times F}$ with F denoting the number of features per node, to capture the detailed characteristics of each node in the static graph G^t .

Given the adjacency matrices and feature matrices from previous T time slices, denoted as $\{A^1, A^2, \dots, A^T\}$ and $\{H^1, H^2, \dots, H^T\}$, respectively, the objective of the temporal

link prediction task is to forecast the network topology for the next time slice $T + 1$. This can be mathematically formulated as

$$\hat{A}^{T+1} = f((A^1, H^1), (A^2, H^2), \dots, (A^T, H^T)) \quad (1)$$

where $f(\cdot)$ is the predictive model we intend to establish in this article and $\hat{A}^{T+1}$ signifies the anticipated adjacency matrix for the time slice $T + 1$.

B. Model Development

The comprehensive architecture of the HENN model is illustrated in Fig. 1. The model is systematically divided into five parts. In part 1, we focus on feature engineering and the construction of temporal networks. In part 2, we utilize GAT to learn positional embeddings for each node within the static networks, capturing their structural relationships. In part 3, vector representations of edges in each static graph are generated based on the positional embeddings from part 2. In part 4, we employ TT to integrate the learned latent features with temporal dynamics, effectively capturing the evolution of the network over time. In part 5, we use MLP to forecast the trade network of next year by the output of part 4. The workflow of HENN is as follows.

- 1) Inputting the adjacency matrices and feature matrices of previous T time slices $\{A^1, A^2, \dots, A^T\}$ and $\{H^1, H^2, \dots, H^T\}$.
- 2) Selecting features and constructing temporal networks.
- 3) Employing GAT to derive vector representations for each node within the static graph.
- 4) Constructing vector representations for edges in each static graph.
- 5) Utilizing TT to obtain vector representations of edges for time slice $T + 1$.
- 6) Using MLP to forecast the $\hat{A}^{T+1}$ based on the vector representations of edges for time slice $T + 1$.

1) Feature Engineering (Part 1): The missing values in node features str handled as follows: If a specific feature is never available for a given country throughout the entire time span, the corresponding feature of that country is set to 0. This provides a consistent unknown encoding while keeping a fixed input dimensionality required by our end-to-end model. If a feature is available for a country in most years but missing in a subset of years, the “country–feature” values are treated as a univariate yearly time series and ARIMA-based imputation is used to fill the missing entries, so that the temporal trend and variability are better preserved compared to simplistic filling strategies. After imputing missing values, the Pearson correlation coefficient (PCC) is adopted as a preliminary feature screening mechanism to remove noise irrelevant to predictive modeling, thereby improving curve-fitting accuracy in subsequent stages. Alternative methods are less suitable in our setting: mutual information typically requires reliable density estimation and becomes unstable when applied to short yearly time series; LASSO is a supervised sparse selector that depends on a clearly defined response variable and linear modeling assumptions; and the F-test relies on univariate significance testing, which does not naturally extend to multiclass or structured prediction scenarios. In contrast, PCC offers a lightweight, interpretable, and statistically stable criterion for measuring linear dependency, enabling effective filtering of weakly correlated or irrelevant features while preserving the most influential variables for downstream modeling. Let $X = [X_1, X_2, \dots, X_n]$ and $Y = [Y_1, Y_2, \dots, Y_n]$ be data samples of two consecutive variables, the PCC of X and Y is calculated as follows:

$$\text{PCC}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (2)$$

where X_i and Y_i denote the i th sample values of variables X and Y , respectively, and $\bar{X}$ and $\bar{Y}$ represent the mean values of X and Y . The PCC spans the range of $[-1, 1]$, with 0 signifying no correlation, 1 indicating perfect positive correlation, and -1 denoting perfect negative correlation. This feature selection process not only improves the efficiency of the modeling process but also enhances the interpretability and predictive power of the resulting model by emphasizing variables that are most relevant to the trend component. In our experiment, we set a parameter threshold for the selection of PCC, and we only retain features with PCC greater than threshold. By employing the PCC algorithm, we have retained F' features for our model.

2) Extracting Features by GAT (Part 2): We employ GAT to capture the local topological structure of individual graph snapshots within a dynamic network. Unlike traditional GNNs, such as GCNs, which aggregate neighbor information using static weights derived from the adjacency matrix and treat all neighbors uniformly, GAT introduces an attention mechanism. This mechanism dynamically computes attention coefficients for each edge, enabling the model to assign different weights to neighbors based on their relevance. As a result, GAT offers greater flexibility and excels at modeling intricate relationships within complex graph structures. Consider a static graph with $|V|$ nodes, each having F' -dimensional features. The graph's structure and node attributes can be expressed with an adjacency matrix $A \in R^{|V| \times |V|}$ and a feature matrix $H \in R^{|V| \times F'}$, where the i th row of H represents the feature vector of node i . First, GAT performs a linear transformation on the features of each

node to generate the input features of each head

$$h_i^k = W^k H_i \quad (3)$$

where $W^k \in R^{D \times F'}$ is the weight matrix of the k th head, $h_i^k \in R^D$ is the features of node i transformed by the k th head. Second, for each edge (i, j) (node i is a neighbor of node j), GAT calculates the attention coefficient and normalizes it

$$e_{ij}^k = \text{LeakyReLU}(\alpha^k \cdot [h_i^k \| h_j^k])$$

$$\alpha_{ij}^k = \frac{\exp(e_{ij}^k)}{\sum_{j' \in N_i} \exp(e_{ij'}^k)} \quad (4)$$

where $\alpha^k \in R^{2D}$ is the attention vector of the k th head and N_i is the neighbor set of node i . Finally, for each node i , GAT aggregates neighbor features weighted by the attention coefficient and merge the outputs of k heads

$$h_i'^k = \sum_{j \in N_i} \alpha_{ij}^k h_j^k$$

$$x_i = \parallel_{k=1}^K h_i'^k \quad (5)$$

where $x_i \in R^{K \cdot D}$ is the final vector representation of node i , which K is the total heads of GAT. With the GAT unit, we can get the vector representation of each node in every static graph $\{X^1, X^2, \dots, X^T\}$ ($X^t \in R^{|V| \times K \cdot D}$) by traversing the adjacency matrices $\{A^1, A^2, \dots, A^T\}$ and feature matrices $\{H^1, H^2, \dots, H^T\}$. To simplify the definition of parameters, we denote $D' = K \cdot D$ there.

3) Constructing Edge Representations (Part 3): To forecast the future network, it is imperative to formulate the vector representation of each edge in every static graph. For instance, consider two nodes, u and v , connected by an edge A_{uv}^t in the time slice t . The vector representation of node u and v are denoted as $X_u^t \in R^{D'}$ and $X_v^t \in R^{D'}$, respectively, thus, the vector representation of edge $e_{uv}^t \in R^{2D'}$ and the vector representation $S^t \in R^{|V|^2 \times 2D'}$ of graph G^t can be constructed as follows:

$$S^t = \{e_{uv}^t | u \in V, v \in V\} = \{X_u^t \oplus X_v^t | u \in V, v \in V\}. \quad (6)$$

Through the operations mentioned above, we can get vector representation $\{S^1, S^2, \dots, S^T\}$ of graph $\{G^1, G^2, \dots, G^T\}$ with $\{X^1, X^2, \dots, X^T\}$.

4) Fusing Temporal Information by TT (Part 4): Building on the results from Part 3, we obtain vector representations of each edge within every static graph. Given the dynamic nature of global natural gas trade, we employ TT to enhance the integration of latent features across different timestamps, effectively capturing temporal dependencies and relationships between edges. TT, a powerful deep learning architecture, excels at modeling sequential and temporal data through its self-attention mechanism. Unlike traditional RNNs, TT processes input data in parallel, enabling efficient handling of long-range dependencies. By leveraging multihead self-attention, it dynamically weighs the importance of different temporal features, facilitating the integration of node representations across time. This makes TT particularly well-suited for our task of modeling the evolving dynamics of natural gas trade networks. The overall operation of TT can be described as follows:

$$H(S, h) = \text{softmax} \left(\frac{(SW_h^Q)(SW_h^K)^T}{\sqrt{d_k}} \right) (SW_h^V) W^O$$

$$H(S) = [H(S, 1) || H(S, 2), \dots || H(S, \text{nhead})]$$

Algorithm 1: Algorithm HENN.

Input : $\{A^1, A^2, \dots, A^T\}$ ($A^t \in \mathbb{R}^{|V| \times |V|}$),
 $\{H^1, H^2, \dots, H^T\}$ ($H^t \in \mathbb{R}^{|V| \times F}$)

Output : $\hat{A}^{T+1}$

- 1 Initialization PCC(threshold);
- 2 update $H^t \in \mathbb{R}^{|V| \times F} \rightarrow H^t \in \mathbb{R}^{|V| \times F'}$ by PCC;
- 3 Initialization TT(heads),MLP(layers);
- 4 $S = []$;
- 5 **for** $t = 1$ **to** $T-1$ **do**
- 6 update X^t with A^t and H^t by GAT;
- 7 $S^t, y_{\text{true}} = [], []$;
- 8 **for** $u = 1$ **to** $|V|$ **do**
- 9 **for** $v = 1$ **to** $|V|$ **do**
- 10 $S^t.\text{append}(X_u^t \oplus X_v^t)$;
- 11 $y_{\text{true}}.\text{append}(A^{t+1}[u][v])$;
- 12 $S.\text{append}(S^t)$;
- 13 update S^{t+1} with S by TT;
- 14 loss=criterion(y_{true} , MLP(S^{t+1}));
- 15 update X^T with A^T and H^T by GAT;
- 16 $S^T = []$;
- 17 **for** $u = 1$ **to** $|V|$ **do**
- 18 **for** $v = 1$ **to** $|V|$ **do**
- 19 $S^T.\text{append}(X_u^T \oplus X_v^T)$;
- 20 $S.\text{append}(S^T)$;
- 21 update S^{T+1} with S by TT;
- 22 $\hat{A}^{T+1} = \text{MLP}(S^{T+1})$;
- 23 **return** $\hat{A}^{T+1}$;

$$\text{FFN}(S) = W_2 \cdot \max(0, W_1 S + b_1) + b_2$$

$$S = \text{FFN}(\text{Norm}(S + H(S))) \quad (7)$$

where nhead means the total heads of TT, $W_h^Q, W_h^K, W_h^V \in \mathbb{R}^{2D' \times d_k}$, $d_k = \frac{2D'}{\text{nhead}}$ are the learnable weight matrices. $W_1, W_2 \in \mathbb{R}^{2D' \times 2D'}$ are the weight matrices of FFN, and b_1, b_2 are the bias vectors. We use TT to get the edge representations S^{T+1} with $\{S^1, S^2, \dots, S^T\}$.

5) Forecasting Future Network by MLP (Part 5): After integrating the learned latent features with temporal information, we employ MLP to execute our predictions. In our model, the formula for MLP in the time slice t is delineated as follows:

$$\hat{A}^t = \sigma(S^t \cdot W^q + b^q) \quad (8)$$

where W signifies the trainable weight matrix, b indicates the bias vector, and q denotes the total number of fully connected layer in our model. The rectified linear unit (ReLU) activation function is employed across these layers, applying element-wise rectification to introduce nonlinearity into the network. The formula for the loss function in our model is described as follows:

$$y = \{(u, v, A_{uv}^t) | u \in V, v \in V\}$$

$$\text{loss} = -\frac{1}{|V|^2} \sum_{i=1}^{|V|^2} \frac{e^{\hat{A}_{i,y_i}^t}}{\sum_{k=1}^3 e^{\hat{A}_{i,k}^t}}. \quad (9)$$

We use the loss function to train the parameters in GAT, TT, and MLP by end-to-end learning. Finally, we use MLP for the forecast of $\hat{A}^{T+1}$ with the vector representation S^{T+1} .

C. Algorithm Design

Building upon the preceding analysis, we developed the HENN model to effectively forecast global natural gas trade.

Initially, we use PCC to select features for our model. For each timestep, adjacency matrix and feature matrix are input into GAT layer, which employs an attention mechanism to generate enriched node embeddings. Next, the node embeddings in each timestep are used to construct edge embeddings by concatenating the features of source and target nodes. Then, these edge embeddings are processed by TT, which captures temporal dependencies across the sequence. Finally, the edge embeddings are fed into MLP, which outputs predictions across three classification classes. The detailed workflow of HENN is outlined in Algorithm 1, presented in pseudocode below. From the pseudocode, we deduce that the model HENN's time complexity is $\Theta(T \times (N \times T \times F^2 + |V|^2 \times F^2))$. Besides, the space complexity of HENN is $\Theta(|V| \times T \times F)$. Our training duration is about five minutes when epoch is set to 200 and early-stop is set to 30, and GPU configuration is 12th Gen Intel Core i7-12700.

V. EXPERIMENTS AND ANALYSIS

In this section, we initially describe the experimental framework, including datasets, baseline models, evaluation metrics, and hyperparameter settings. Subsequently, we present a detailed comparison of the proposed HENN model against other approaches. After that, a comprehensive ablation study is conducted to evaluate the individual contributions of HENN's architectural components. Then, we include a case study to demonstrate the interpretability of HENN using attention-based visualizations. In the end, we conduct parameter sensitivity analyses to assess the impact of varying hyperparameters on the HENN's performance. Besides, our experiments will revolve around and answer the following three questions.

RQ1: Can the proposed model HENN be capable of surpassing current advanced models in the forecast of global natural gas trade?

RQ2: How do different parts influence the performance of the model HENN?

RQ3: How do different hyperparameter settings influence the performance of the model HENN?

RQ4: How can the interpretability components of HENN be used to explain model decisions in forecasting national natural gas trade?

A. Experimental Setup

1) Datasets: The dataset employed in this study is sourced from British Petroleum, a globally acclaimed authority in the oil and petrochemical industry, ensuring the reliability and relevance of the data for our analysis. The dataset is publicly accessible via.² The original dataset encompasses diverse energy categories, including oil, coal, electricity, and others. For the purpose of this study, we exclusively focused on natural gas-related data, which was refined to cover a consistent 24-year period from 2000 to 2023. This comprehensive dataset incorporates a rich array of country-specific features, such as natural gas reserves, energy import/export volumes, and key economic indicators.

To adapt the data to our analytical framework, we transformed the original regression task into a classification problem because our primary goal is to forecast the future structure and strength tiers of the trade network, rather than to recover precise

²[Online]. Available: <https://www.bp.com> and <https://www.energyinst.org/statistical-review>

TABLE I
SUMMARY OF COMPARISON MODELS

Model	Description
DySAT [7]	It is an innovative neural architecture that acquires node representations to track the evolving structure of dynamic graphs (WSDM 2020).
TGAT [10]	It aggregates temporal-topological neighborhood features as well as learn the time-feature interactions (ICLR 2020).
TGN [8]	It is a generic, effective framework for deep learning on dynamic graphs represented as sequences of timed events (ICLR 2020).
EvolveGCN [9]	It adopts the GCN model along the temporal dimension without resorting to node embeddings (AAAI 2020).
FastSTLGS [31]	It is an advanced model that autonomously examines topological characteristics within a cohesive framework, effectively modeling the spatio-temporal relationships in Mobile Ad Hoc Networks (Scientific Reports 2022).
GC-LSTM [11]	It is a unique model encapsulating both the local structural attributes of nodes and their interconnections (Applied Intelligence 2022).
FreeDyG [13]	It introduces a module for encoding node interaction frequency, explicitly capturing the shared neighbors and the interaction frequency between node pairs (ICLR 2024).
GCN_MA [32]	It integrates GCN, RNN, and multi-head attention to produce detailed and precise node embedding representations (Scientific Reports 2024).
HENN	Our approach combines GAT, TT, and MLP to forecast global natural gas trade through end-to-end learning.

The bold sentence is to highlight our own proposed model.

continuous volumes for every bilateral edge. The classification criteria are as follows: trade volumes accounting for less than 10% of the total trade volume are classified as level 0, those between 10% and 60% as level 1, and volumes exceeding 60% as level 2. This design is motivated by a combination of statistical characteristics of the data and practical industry considerations. Global natural gas trade volumes exhibit a highly skewed and heavy-tailed distribution, in which a small number of links account for a disproportionately large share of total trade. Under such conditions, directly modeling continuous volumes would make learning overly sensitive to extreme values and cross-year scale variations. Discretizing trade volumes into three tiers alleviates these issues by stabilizing learning targets, reducing the influence of outliers, and mitigating class imbalance, while still preserving meaningful distinctions between low-, medium-, and high-intensity trade relationships. Empirically, the 10% and 60% thresholds provide a reasonable separation of marginal trade links, stable medium-intensity relationships, and dominant strategic corridors observed in the data. From an industry perspective, links contributing less than 10% typically correspond to occasional or low-impact trade flows, whereas links exceeding 60% represent core bilateral dependencies that are critical for energy security and long-term strategic planning. Therefore, the adopted thresholds align both with the statistical distribution of trade volumes and with practical interpretations used in energy trade analysis, making them suitable for robust and interpretable network forecasting. To enhance the robustness of our comparative experiments and ablation studies, we established three test sets covering the periods 2000, 2001,..., 2021, 2000, 2001,..., 2022, and 2000, 2001,..., 2023. Detailed descriptions of each country with its features are shown in Table II and the statistical properties are summarized in Table III.

2) Baseline Methods: The HENN model is evaluated against eight prominent advanced models, including three GCN-based approaches (GC-LSTM, GCN_MA, and EvolveGCN) and two GAT-based models (DySAT and TGAT), alongside other

TABLE II
DETAILED DESCRIPTIONS OF EACH COUNTRY WITH ITS FEATURES

Country	Features
U.S.	Natural gas production, Proven reserve of natural gas, Biofuel generation, Coal consumption, etc.
Oman	Natural gas consumption, Primary energy consumption, Carbon dioxide from Methane and process, etc.
Qatar	Natural gas production, Natural gas consumption, Primary energy consumption, etc.
:	:
Canada	Carbon dioxide from Burning, Coal production, etc.
China	Coal production, Renewable energy, Carbon dioxide from Burning, etc.
Korea	Primary energy consumption, Biofuel consumption, Biofuel generation, etc.

TABLE III
STATISTICS OF THE STUDIED DATASETS

Year	Countries	Edges	Max-Volume	Total-Volume
2000	24	41	24.25	136.96
2001	27	44	22.74	142.95
2002	27	48	23.4	149.99
2003	28	43	24.05	168.84
:	:	:	:	:
2020	49	178	40.6	487.9
2021	49	163	43.6	516.2
2022	49	180	41.9	542.4
2023	43	173	37.5	549.2

competitive frameworks. This comparative analysis highlights HENN's performance relative to these established methods in dynamic graph analysis. Table I offers detailed summaries of each competing model, outlining their architectural foundations and key characteristics to provide context for the evaluation.

3) Evaluation Metrics: Building on prior research, we set seven well-established evaluation metrics to assess model performance: accuracy, precision, recall, Macro-F1, Micro-F1, area under the ROC curve (AUC), and area under the precision-recall curve (AUPR). To determine these metrics, the construction of a confusion matrix is essential. In a confusion matrix, we identify true positives (TP) as instances with positive actual and predicted values, false positives (FP) as those with positive actual but incorrect predictions, false negatives (FN) as those with negative actual but incorrect predictions, and true negatives (TN) as instances with negative actual and predicted values. The calculation formulas of four indicators are as follows:

$$\begin{aligned}
 \text{Accuracy} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \\
 \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\
 \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\
 \text{Macro-F1} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (10)
 \end{aligned}$$

4) Hyperparameters: In our experiment, we utilized four hyperparameters: hidden_dim, heads, layers, and threshold. hidden_dim defines the dimensionality of the vector representation for each node in our model. Heads indicates the number of attention heads in the TT architecture. Layers specifies the number of MLP layers, which varies from 1 to 5 in our configuration.

TABLE IV
MODEL PERFORMANCE COMPARISON ACROSS YEARS

Year	Metric	DySAT	FastSTLSG	FreeDyG	GC-LSTM	GCN_MA	TGN	EvolveGCN	TGAT	HENN
2021	Accuracy	0.677±.012	0.596±.015	0.548±.020	0.662±.018	0.657±.016	0.634±.014	0.649±.017	0.611±.013	0.712 ±.008
	Precision	0.657±.010	0.584±.019	0.488±.025	0.646±.011	0.608±.017	0.546±.022	0.548±.021	0.550±.015	0.698 ±.009
	Recall	0.534±.030	0.581±.028	0.488±.035	0.490±.026	0.601±.024	0.459±.032	0.506±.029	0.483±.027	0.732 ±.007
	Macro-F1	0.589±.015	0.582±.018	0.488±.022	0.557±.016	0.604±.014	0.498±.020	0.526±.019	0.514±.017	0.715 ±.006
	Micro-F1	0.677±.012	0.596±.015	0.548±.020	0.662±.018	0.657±.016	0.634±.014	0.649±.017	0.611±.013	0.712 ±.008
	AUC	0.710±.009	0.774±.007	0.706±.011	0.699±.010	0.795±.006	0.765±.008	0.743±.009	0.721±.010	0.861 ±.005
2022	AUPR	0.548±.012	0.607±.010	0.516±.015	0.538±.011	0.637±.009	0.576±.013	0.563±.014	0.521±.012	0.730 ±.006
	Accuracy	0.619±.018	0.619±.016	0.503±.022	0.583±.017	0.702±.010	0.621±.015	0.629±.019	0.566±.020	0.785 ±.007
	Precision	0.602±.015	0.619±.014	0.475±.025	0.587±.016	0.661±.011	0.666±.010	0.602±.017	0.681±.009	0.754 ±.008
	Recall	0.505±.030	0.636±.025	0.492±.038	0.531±.028	0.665±.020	0.497±.035	0.540±.032	0.475±.036	0.758 ±.006
	Macro-F1	0.549±.018	0.627±.016	0.483±.024	0.558±.017	0.663±.012	0.569±.022	0.569±.021	0.559±.019	0.756 ±.007
	Micro-F1	0.619±.018	0.619±.016	0.503±.022	0.583±.017	0.702±.010	0.621±.015	0.629±.019	0.566±.020	0.785 ±.007
2023	AUC	0.736±.008	0.696±.010	0.735±.009	0.707±.011	0.855±.005	0.771±.007	0.765±.008	0.667±.012	0.891 ±.005
	AUPR	0.575±.015	0.654±.012	0.531±.018	0.539±.014	0.726±.008	0.598±.016	0.574±.017	0.513±.019	0.786 ±.006
	Accuracy	0.634±.020	0.662±.018	0.583±.025	0.599±.019	0.705±.012	0.634±.017	0.647±.020	0.583±.018	0.808 ±.009
	Precision	0.647±.016	0.631±.015	0.539±.022	0.619±.017	0.662±.013	0.564±.020	0.558±.021	0.502±.024	0.781 ±.008
	Recall	0.483±.035	0.666±.028	0.557±.040	0.529±.032	0.667±.022	0.520±.038	0.539±.035	0.580±.030	0.792 ±.007
	Macro-F1	0.553±.020	0.648±.018	0.528±.025	0.570±.019	0.665±.014	0.541±.022	0.548±.021	0.538±.020	0.786 ±.008
Statistic	Micro-F1	0.634±.020	0.662±.018	0.583±.025	0.599±.019	0.705±.012	0.634±.017	0.647±.020	0.583±.018	0.808 ±.009
	AUC	0.741±.007	0.829±.006	0.767±.008	0.734±.009	0.862±.005	0.782±.006	0.754±.008	0.770±.007	0.898 ±.005
	AUPR	0.562±.018	0.682±.015	0.606±.020	0.564±.017	0.743±.009	0.593±.018	0.667±.016	0.581±.020	0.789 ±.007
Statistic	Loss/Win	0/21	0/21	0/21	0/21	0/21	0/21	0/21	0/21	0/168
	<i>p</i> -value	0.0000009	0.0000009	0.0000009	0.0000009	0.0000009	0.0000009	0.0000009	0.0000009	-
	F-rank	5.64	3.88	7.83	6.33	2.29	6	5.24	6.79	1

The bold values denote the best performance among all the involved methods.

Threshold sets the criterion for feature selection based on the PCC. For instance, setting the threshold to 0.1 means that only features with PCC greater than 0.1 can be selected as model inputs.

B. Comparison Results (RQ.1)

Table IV presents the evaluation metrics for all models across various testing sets. To further explore these results, we conducted statistical analyses of loss/win, Wilcoxon Signed-ranks Test (WST), and Friedman Test. The loss/win metric measures how often HENN outperforms or underperforms other models in terms of seven evaluation metrics across all test datasets. WST assesses whether HENN's prediction accuracy significantly surpasses that of competing models, with the *p*-value indicating the degree of statistical significance. In addition, Friedman Test evaluates the performance of multiple models across several datasets simultaneously, using the F-rank value as an indicator, where a lower F-rank signifies superior predictive performance. The results for loss/win, *p*-value, and F-rank are reported in the penultimate, antepenultimate, and final rows of Table IV, respectively. According to Table IV, the HENN model demonstrates outstanding performance across all evaluated metrics, consistently outperforming baseline models in global natural gas trade forecasting. For accuracy, HENN achieves the highest scores across all years, outperforming the second-best model by 4%–10%. This superior accuracy underscores its robust generalization capabilities when handling dynamic graph data, effectively capturing evolving trade patterns. For precision and recall, HENN maintains the best balance between precision and recall, while other models show significant tradeoffs. This highlights its ability to minimize both FP and FN. For AUC and AUPR, HENN consistently achieves the highest AUC and AUPR, outperforming the second-best models by 4%–6% and 4%–10%. This confirms its robustness in handling noisy data and

maintaining high precision-recall tradeoffs under class imbalance. Table IV indicates that HENN's margin over the strongest baseline becomes larger on discrete decision metrics for the 2023 test set. We attribute this to (i) a more concentrated and structurally sensitive network in 2023, where the number of countries decreases while the total trade volume increases, implying that major links carry more weight and structural shifts become more impactful and (ii) HENN's end-to-end spatial-temporal integration, where GAT extracts structure-aware representations and TT captures long-range yearly dependencies, jointly optimized toward next-year network forecasting. This design is particularly robust under complex and uncertain drivers emphasized in our motivation. Three key observations from statistic in Table IV include the following.

1) HENN exhibits unparalleled estimation precision across three testing sets, achieving an impressive loss/win ratio of 168 wins to 0 losses in head-to-head comparisons with each competing model.

2) The *p*-values are all significantly below 0.05, indicating that HENN's prediction accuracy is markedly higher than that of all other models under comparison.

3) HENN attains the lowest F-rank score among all evaluated models, providing further evidence of its dominant predictive accuracy across the test datasets.

The key observations mentioned above highlight HENN's superior performance in global natural gas trade forecast, with a perfect 168-0 loss/win record, statistically significant *p*-values below 0.05, and the lowest F-rank score, demonstrating its unmatched accuracy, robustness, and ability to outperform competing models across all evaluation metrics.

C. Ablation Experiments (RQ.2)

To evaluate the individual contributions of HENN's architectural components, we conducted systematic ablation experiments comparing the complete model against three degraded

TABLE V
ABLATION EXPERIMENT RESULTS

Year	Metric	No-Feature	No-GAT	No-TT	Original
2021	Accuracy	0.5884	0.6944	0.7045	0.7121
	Precision	0.1961	0.6142	0.6943	0.6983
	Recall	0.3333	0.6178	0.6359	0.7316
	F1-score	0.247	0.6159	0.6566	0.7145
	AUC	0.5	0.8275	0.8128	0.861
	AUPR	0.6667	0.6713	0.6646	0.7299
2022	Accuracy	0.5455	0.7172	0.7197	0.7854
	Precision	0.1818	0.6809	0.6868	0.7541
	Recall	0.3333	0.696	0.6806	0.7577
	F1-score	0.2353	0.6875	0.6776	0.7554
	AUC	0.5	0.8471	0.8558	0.891
	AUPR	0.6667	0.7232	0.7324	0.7855
2023	Accuracy	0.5631	0.7298	0.75	0.8081
	Precision	0.1877	0.6901	0.7254	0.7811
	Recall	0.3333	0.718	0.7161	0.7912
	F1-score	0.2402	0.7018	0.7146	0.7861
	AUC	0.5	0.8635	0.852	0.8976
	AUPR	0.6667	0.7537	0.7463	0.7885
Statistic	Loss/Win	0/18	0/18	0/18	0/54
	<i>p</i> -value	0.000008	0.000008	0.000008	-
	F-rank	3.94	2.61	2.44	1

The bold values denote the best performance among all the involved methods.

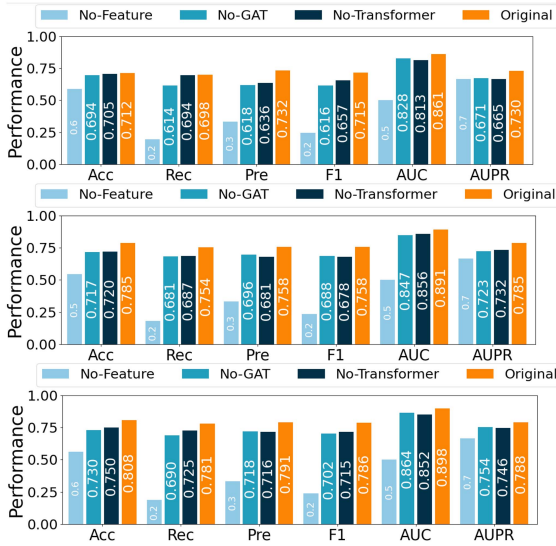


Fig. 2. Visualization of ablation experiments.

variants: No-Feature, No-GAT, and No-TT. The results are presented in Table V. The “Original” configuration refers to the full HENN model with all constituent base models. The “No-Feature” variant excludes all node attributes, such as natural gas reserves, energy imports and exports, economy, and population. The “No-GAT” variant omits the GAT module, while the “No-TT” variant excludes the TT module. To clearly illustrate the performance differences between the full HENN model and these degraded variants, we visualized the results in Fig. 2, highlighting each component’s role. Four key observations emerge from Table V and Fig. 2.

- 1) Node features serve as the cornerstone of the HENN model, underpinning its ability to discern intricate structural and semantic patterns within dynamic graphs. predicting trade-volume levels depends not only on connectivity but also strongly on country capacity and demand drivers, which are exactly what our node features represent. Once these semantic signals are removed, the model

is forced to infer trade strength purely from topology, which is insufficient for this edge-level volume-tier prediction task and, therefore, leads to the most pronounced performance degradation, underscoring their indispensable role in establishing a robust foundation for effective graph representation learning.

- 2) The GAT module plays a pivotal role in spatial feature aggregation by dynamically weighting influential nodes based on their contextual relevance. Ablation studies reveal consistent reductions in recall and AUC when the GAT module is omitted, affirming its critical contribution to enhancing the model’s ability to capture localized relational dependencies.
- 3) The TT module, though less pronounced in its impact, is essential for robust temporal modeling. Its absence leads to a notable decline in performance, with reductions in F1-score ranging from 4% to 8% and AUPR dropping by approximately 5%. These findings highlight the TT’s role in effectively capturing long-range temporal dependencies, thereby enriching the model’s dynamic analysis capabilities.
- 4) The integrated HENN model, which synergistically combines node features, GAT-based attention mechanisms, and TT-driven temporal reasoning, achieves optimal performance across diverse test sets. Statistical validation confirms its superiority over ablated variants, demonstrating that the cohesive interplay of these components is crucial for delivering advanced results in dynamic graph analysis.

In summary, the HENN model’s exceptional efficacy is driven by the complementary roles of its core components. Node features lay the groundwork for robust pattern recognition, the GAT module refines spatial interactions, and the TT module enhances temporal coherence. As a result, the above three modules are all indispensable for our model.

D. Sensitivity Analyses of Hyperparameters (RQ.3)

The systematic optimization of four critical hyperparameters significantly contributes to our model’s enhanced performance, as quantitatively demonstrated in Fig. 3 through comprehensive ablation studies. Fig. 3(a) reveals that 64-D hidden representations achieve optimal accuracy by maintaining an effective balance between computational tractability and feature expressiveness. Notably, insufficient dimensionality compromises feature learning capacity, while higher dimensions escalate computational overhead without measurable performance benefits. Fig. 3(b) examines multilayer perceptron depth variations, where a triple-layer architecture maximizes both AUC and AUPR metrics. Contrary to expectations, deeper architectures degrade performance sharply, indicating overfitting. As shown in (c), the threshold of PCC exhibits nonlinear impacts on model accuracy, with optimal performance observed within the 0.3–0.5 interval, while extreme thresholds, respectively, introduce redundant noise or excessive feature elimination. Finally, (d) investigates multihead attention mechanisms in TT layers, demonstrating that 8-head configurations optimally capture diverse feature interactions. These systematic investigations underscore

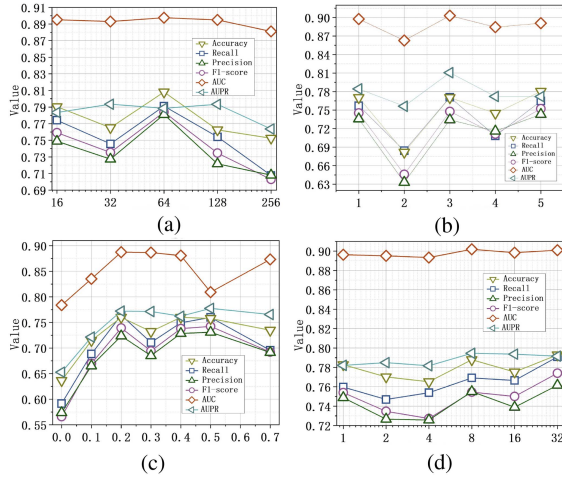


Fig. 3. Results of Different Hyperparameters. (a) Hidden_dim. (b) Layers of MLP. (c) Threshold of PCC. (d) Heads of TT.

the critical importance of meticulous hyperparameter configuration, revealing that both under specified and over engineered parameterizations can degrade model efficacy across different architectural components. As evidenced by comprehensive ablation studies, the systematic optimization of hyperparameters is crucial for maximizing our model's performance.

E. Case Study (RQ.4)

HENN can provide interpretable evidence about which factors are most associated with predicted changes in trade volume levels and thus help explain trade fluctuations from a model-driven perspective. HENN can make interpretability explicit along the following three complementary axes.

- 1) HENN pipeline already performs PCC-based feature screening, which offers an initial, transparent signal of which covariates are most correlated with trade-related targets.
- 2) The spatial encoder produces attention weights over neighboring countries, which can be interpreted as identifying which trade partners/structural neighbors contribute most to a country's representation at a given time.
- 3) The TT provides self-attention over historical snapshots, which can indicate which past years/time segments are most influential when forecasting the next-year trade network.

This helps distinguish whether predictions are driven more by recent shocks versus longer-term trends. Based on the above interpretability design, we conduct a case-study-based analysis to make these mechanisms concrete and observable. Specifically, we focus on representative countries and visualize how HENN explains trade dynamics from three complementary perspectives: feature relevance, spatial dependency, and temporal influence.

First, for each selected country, we examine feature-level relevance by leveraging the PCC screening already embedded in the HENN pipeline. PCC scores are computed between candidate country-level covariates and the corresponding trade-related targets, allowing us to identify the top-ranked features that are most strongly associated with predicted trade volume levels. This step provides an initial and transparent indication of which economic

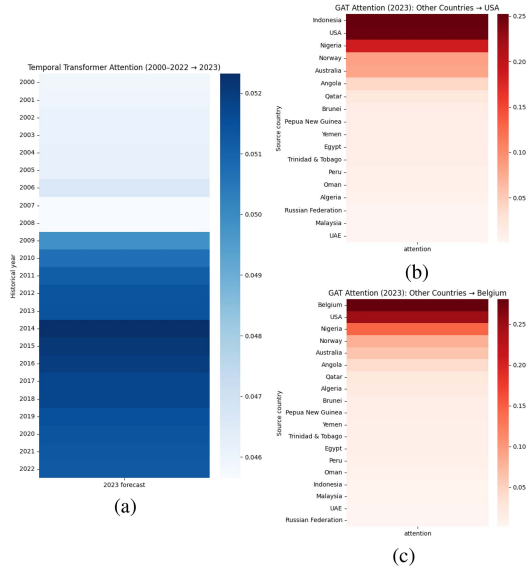


Fig. 4. Visualization examples of attention mechanisms. (a) TT attention. (b) GAT attention of USA. (c) GAT attention of Belgium.

or energy-related factors play a dominant role in shaping the model's forecasts. For China, the top-ranked features identified by PCC include nuclear energy generation (TWh), multiple measures of natural gas consumption (Exajoules, Bcf, and Bcm), as well as renewable energy generation from geothermal, biomass, and other sources. This pattern suggests that the model associates China's trade dynamics not only with the scale of domestic gas demand, but also with the broader structure of its energy mix and the interaction between gas, nuclear, and renewable energy sources. In contrast, for Japan, the most influential features are dominated by nuclear energy and renewable energy indicators, including nuclear energy generation and consumption, renewable energy consumption, and overall electricity generation. Compared with China, direct gas consumption variables play a less prominent role, reflecting Japan's stronger dependence on energy imports and the closer coupling between its gas trade patterns and the configuration of its power generation portfolio.

Next, we analyze spatial dependencies by extracting attention weights from the GAT encoder for the 2023 trade network. For each target country (e.g., the United States and Belgium), we visualize the attention coefficients assigned to its neighboring countries using heatmaps. These attention scores reflect the relative contribution of different trade partners or structurally connected countries to the target country's representation, thereby highlighting key bilateral relationships emphasized by the model in the prediction process. As shown in Fig. 4(b) and (c), the GAT attention distributions for the United States and Belgium in 2023 emphasize a small number of major exporting countries, such as the U.S., Nigeria, Norway, and Australia, while assigning substantially lower weights to peripheral suppliers. This sparse and asymmetric attention pattern is consistent with real-world trade structures, where a limited set of key partners dominates national gas imports.

Finally, we investigate temporal influence by visualizing self-attention weights from the TT when forecasting the 2023 trade network. Specifically, we plot the attention assigned to each historical snapshot from 2000 to 2022 with respect to the 2023 prediction. This visualization reveals which past periods exert

the greatest influence on current forecasts, helping distinguish whether model predictions are driven primarily by recent shocks or by longer-term historical trends. Fig. 4(a) further indicates that the TT assigns higher attention weights to recent years, particularly the post-2009 period, suggesting that the model places greater emphasis on more recent structural changes and market conditions when forecasting the 2023 trade network.

VI. CONCLUSION

This article proposes a novel HENN model for forecasting global natural gas trade. HENN comprises three components: GAT, TT, and MLP. First, GAT learns the latent features of annual trade network among various countries year by year. Second, TT integrates these annual latent features with temporal information to forecast the trade network's representations for the upcoming year. Third, MLP employs the forecasted representations to forecast the trade network of next year. The three components are trained by end-to-end learning, ensuring the feature extraction and the integration of temporal information are both comprehensive and optimal for forecasting. To evaluate the proposed HENN model, extensive experiments are conducted on global natural gas trade data, including comparative experiments, ablation experiments, and parameter sensitivity experiments. The results mentioned above demonstrate that HENN not only achieves high levels of accuracy but also markedly outperforms current advanced models in forecasting the global natural gas trade.

A key limitation of HENN is that several important hyperparameters still require manual tuning, including the PCC screening threshold as well as architectural choices such as the number of layers and attention heads. To reduce reliance on manual trial-and-error and improve generalization, we plan to make these hyperparameters adaptive within the end-to-end training pipeline. Specifically, we will replace fixed threshold-based screening with learnable feature gating/sparsity mechanisms, allowing the model to suppress noninformative features during training automatically. We will explore automated model selection strategies to adaptively choose the appropriate network depth and attention-head configurations. Besides, we will explore coupling a data-driven forecasting framework with fuzzy multicriteria decision-making paradigms to better support risk-oriented interpretation. In particular, inspired by Kan et al. [35], we plan to investigate whether similar grey-VIKOR style fusion can be used to (i) incorporate expert preferences and subjective uncertainty and (ii) aggregate multilevel indicators derived from both observed attributes and model-predicted network changes. This integration would provide a more transparent and uncertainty-aware decision layer on top of the forecasted trade network, enabling scenario-based ranking and prioritization while retaining the predictive strength of the proposed end-to-end model.

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